

# Mai Hai Dang

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*Embedded Software Engineer with 2 years hands-on experience developing AUTOSAR-based automotive software for EV powertrain and IVI systems. Experienced in diagnostic communication (UDS/DCM), CAN/LIN/Ethernet networks, software integration, testing, and ASPICE-compliant development processes. Strong foundation in embedded C, software architecture, debugging, and automotive standards including ASPICE and MISRA C.*

## Experience

### FPT Software

Dec 2024 – Present

*Embedded Software Engineer*

Hoa Lac Hi-Tech Park, Thach That, Hanoi, Vietnam

- Architected and implemented application-level IVI features — navigation, HUD, and Meter Display — across production automotive platforms
- Debugged, tested, and validated software on real hardware, debug boards, and simulators to improve stability and reduce defect rates. Tools used: Vector CANalyzer, Vector CANoe, and Green Hills MULTI.
- Delivered full ASPICE SWE.1–SWE.5 documentation suite: requirements specifications, software architecture, detailed design, traceability matrices, and verification reports.
- Integrated and debugged CAN, Ethernet, and LIN communication modules for ECU-level feature development and cross-module communication.
- Collaborated within Agile/Scrum teams to support software releases and continuous integration activities

### Bosch Global Software Technologies Co., Ltd (BGSV)

Apr 2024 – Sep 2024

*Embedded Software Engineer*

29 Lieu Giai, Ba Dinh, Hanoi

- Projects: Vinfast VF3/VF8/VF9 EV Powertrain System.
- Developed AUTOSAR R4.4.0-compliant embedded C software for UDS/DCM diagnostic services — DID RDBI/WDBI, DTC management, session control, and security access — over CAN bus for the Vinfast EV Powertrain System
- Configured CAN gateway routing (ComM, CanIf, CanSM, PduR) to enable reliable cross-ECU communication with chassis and body control modules, resolving inter-ECU integration bottlenecks
- Creating automated functional software testing packages using Tracetonic ecu.test, ETAS INCA, and Vector CANalyzer. Conducting unit testing with Testwell CTC++. Performing integration test for new software components.
- Following strictly ASPICE for SWE process group. Adhering to ISO 14229-1:2020, ISO 11898-1:2024, ISO 26262-2:2018, ISO 15765-2:2024, MISRA C:2024 and ASAM AE MDX, MCD-2 MC.

### USTH Learning Support

Dec 2021 – Oct 2023

*Operations Manager and ICT Specialist*

18 Hoang Quoc Viet, Cau Giay, Hanoi

- Effectively managing a team of 20 individuals with diverse scientific backgrounds. Securing recognition as the university's top contributor to student academic success.
- Offering comprehensive academic support to ICT students, focusing on core concepts like data structures and algorithms, OOP, software engineering, networking, web development, AI, and more.

## Education

### University of Science and Technology of Hanoi (USTH)

Oct 2021 – Dec 2024

Bachelor Information and Communication Technology

GPA: 3.5

18 Hoang Quoc Viet, Cau Giay, Hanoi

## Awards

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| <b>Bachelor's Degree with Distinction</b>                                                        | Dec 2024 |
| Bachelor's degree with Distinction in Information and Communication Technology.                  | USTH     |
| <b>Internship Scholarship</b>                                                                    | Oct 2023 |
| Level: A2, second-highest level. Top 6 percent of the university.                                | USTH     |
| <b>Merit Scholarship B1</b>                                                                      | Oct 2022 |
| Level A5. Top 10 percent of the university's General Education Department.                       | USTH     |
| <b>Merit Scholarship B2</b>                                                                      | Oct 2023 |
| Level A5. Top 8 percent of the university's Information and Communication Technology Department. | USTH     |

## Projects

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| <b>Embedded Software for Control and Communication in Powertrain System</b>                                                                                                                                                                                                                                                        |
| <i>C, RTA-CAR</i>                                                                                                                                                                                                                                                                                                                  |
| Configured AUTOSAR BSW modules (ComM, CanIf, CanSM, PduR) to enable CAN communication; integrated UDS diagnostic services via the DCM module; managed non-volatile ECU data with NvM Manager. Implemented application-layer logic for DID data processing supporting RDBI and WDBI UDS services in a production EV powertrain ECU. |
| <b>Web Application for Model Management for Crop Nutrition</b>                                                                                                                                                                                                                                                                     |
| / <i>React, Typescript, Vite, Tailwind</i>                                                                                                                                                                                                                                                                                         |
| <a href="https://github.com/ChloroProject-2023/dashboard-UI">github.com/ChloroProject-2023/dashboard-UI</a>                                                                                                                                                                                                                        |
| Online platform interface for bioinformatic scientists to share machine learning models and datasets for predicting NPK values for rice crop using hyperspectral images and reflectance data. Support train and inference directly on the website.                                                                                 |
| <b>Taxi Information Management System</b>                                                                                                                                                                                                                                                                                          |
| <i>Python</i>                                                                                                                                                                                                                                                                                                                      |
| <a href="https://github.com/Vivarium69420/Taxi-Information-Management-System">github.com/Vivarium69420/Taxi-Information-Management-System</a>                                                                                                                                                                                      |
| A straightforward system for management of customers, invoices, vehicles, and taxi drivers information.                                                                                                                                                                                                                            |

## Skills

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| <b>Programming Language:</b> Java, Python, C/ C++, JavaScript/ TypeScript, Bash |
| <b>Frameworks:</b> AUTOSAR, React, Android SDK, Scikit-learn                    |
| <b>Databases:</b> MariaDB, MySQL, MongoDB                                       |
| <b>Natural Languages:</b> English (B2), French (A2), Vietnamese (Native)        |